

SHUAIKE SHEN

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EDUCATION

Carnegie Mellon University
Ph.D

Pittsburgh, PA, U.S.
Aug. 2025 - Present

Zhejiang University
B.Eng in Computer Science and Technology (Honors)

Hangzhou, Zhejiang, China
Sept. 2021 - June. 2025

- Mixed Class, [Chu Kochen Honors College](#), GPA: 3.83/4.00

RESEARCH INTEREST

I am deeply interested in developing AI and machine learning methods for scientific discovery, particularly in biomedical applications. My research focuses on protein design and flow-matching models. I have also worked on topics related to scientific large language models, reinforcement learning, and crystal structure modeling.

RESEARCH EXPERIENCE

Research Assistant
Advisor: [Prof. Jian Ma](#)

Carnegie Mellon University
Nov. 2025 - Present

- Working on advanced AI agents to accelerate scientific discovery, with a specialized focus on genomic and cellular analysis.

Research Assistant
Advisor: [Prof. Olexandr Isayev](#) & [Prof. Lei Li](#)

Carnegie Mellon University
Aug. 2025 - Nov. 2025

- Working on low energy conformer generation and protein complex using diffusion models.

Research Intern
Advisor: [Dr. Biqing Qi](#) & [Dr. Yuqiang Li](#)

Shanghai AI Lab
May. 2025 - Aug. 2025

- Working on large language models for molecular spectra and developing molecular foundation models to bridge spectral data with molecular properties and structures.

Research Intern
Advisor: [Prof. Ge Liu](#) & [Prof. Nicholas Ching Hai Wu](#)

University of Illinois Urbana-Champaign
June. 2024 - Mar. 2025

- Working on latent flow matching model for protein sequence and structure co-design and developed online reinforcement algorithms for flow matching models.

Research Assistant
Advisor: [Prof. Chunhua Shen](#) & [Prof. Hao Chen](#)

Zhejiang University
Oct. 2022 - May. 2024

- Working on protein engineering and crystal structure modeling. I developed vector field protein encoder, multi-motif scaffolding method, RNA inverse folding paradigm, binding energy prediction model and crystal denoising pretraining framework.

PUBLICATIONS

* denotes equal contribution and † denotes corresponding author.

AgentCo-op: Retrieval-Based Synthesis of Interoperable Multi-Agent Workflows

May. 2026

- Shuaike Shen^{*}, Wenduo Cheng^{*}, Shike Wang, Mingqian Ma, Jian Ma[†]
- Under review, [Paper](#), [Code](#), [Project Page](#)

Can Agents Price a Reaction? Evaluating LLMs on Chemical Cost Reasoning

May. 2026

- Yuyang Wu^{*}, Yue Huang^{*}, Shuaike Shen, Xujian Wang, Shuhao Zhang, Qiyao Xue, Weichen Liu, Runtian Gao, Jian Ma, Xiangliang Zhang, Olexandr Isayev[†]
- Under review, [Paper](#)

SkillFoundry: Building Self-Evolving Agent Skill Libraries from Heterogeneous Scientific Resources	Apr. 2026
• Shuaike Shen[*], Wenduo Cheng[*], Mingqian Ma, Alistair Turcan, Martin JinYE Zhang, Jian Ma[†] • Under review, Paper, Code, Project Page	
MolAct: An Agentic RL Framework for Molecular Editing and Property Optimization	Dec. 2025
• Zhuo Yang[*], Yeyun Chen[*], Jiaqing Xie, Ben Gao, Shuaike Shen, Wanhao Liu, Liujia Yang, Beilun Wang, Tianfan Fu, Yuqiang Li[†] • Technical Report, Paper, Code	
R²Seg: Training-Free OOD Medical Tumor Segmentation via Anatomical Reasoning and Statistical Rejection	Nov. 2025
• Shuaike Shen[*], Ke Liu[*], Jiaqing Xie, Shangde Gao, Chunhua Shen, Ge Liu, Mireia Crispin-Ortuzar, Shangqi Gao[†] • CVPR 2026 Oral, Paper, Code, Project Page	
SpecMol: A Spectroscopy-Grounded Foundation Model for Multi-Task Molecular Learning	Sep. 2025
• Shuaike Shen, Jiaqing Xie, Zhuo Yang, Antong Zhang, Shuzhou Sun, Ben Gao, Tianfan Fu[†], Biqing Qi[†], Yuqiang Li[†] • Under review, Paper, Code	
SpectrumWorld: Artificial Intelligence Foundation for Spectroscopy	Aug. 2025
• Zhuo Yang[*], Jiaqing Xie[*], Shuaike Shen, Daolang Wang, Yeyun Chen, Ben Gao, Shuzhou Sun, Biqing Qi, Dongzhan Zhou, Lei Bai, Linjiang Chen, Shufei Zhang, Jun Jiang[†], Tianfan Fu[†], Yuqiang Li[†] • Under review, Paper	
Intern-S1: A Scientific Multimodal Foundation Model	Aug. 2025
• Intern-S1 team • Technical Report, Paper, Code, Huggingface	
From Sentences to Sequences: Rethinking Languages in Biological System	July. 2025
• Ke Liu[*], Shuaike Shen[*], Hao Chen[†] • Paper, Code	
Online Reward-Weighted Fine-Tuning of Flow Matching with Wasserstein Regularization	Jan. 2025
• Jiajun Fan, Shuaike Shen, Chaoran Cheng, Yuxin Chen, Chumeng Liang, Ge Liu[†] • ICLR 2025, Paper	
A Denoising Pre-training Framework for Accelerating Novel Material Discovery	Dec. 2024
• Shuaike Shen, Ke Liu, Hao Chen[†] • AAAI 2025, Paper	
Training Text-to-Image Flow Models on Self-Generated Data	Nov. 2024
• Jiajun Fan, Shuaike Shen, Chaoran Cheng, Chumeng Liang, Ge Liu[†] • Under review	
Physics-Informed Neural Networks for Unsupervised Binding Energy Prediction	Aug. 2024
• Ke Liu[*], Shuaike Shen[*], Hao Chen[†] • Under review, Manuscript	
Boost Your Crystal Model with Denoising Pre-training	June. 2024
• Shuaike Shen[*], Ke Liu[*], Muzhi Zhu, Hao Chen[†] • ICML 2024 AI for Science Workshop, Paper, Project Page	
FADiff: Floating Anchor Diffusion Model for Multi-motif Scaffolding	May. 2024
• Ke Liu[*], Shuaike Shen[*], Weian Mao[*], Xiaoran Jiao, Zheng Sun, Hao Chen[†], Chunhua Shen[†] • ICML 2024, Paper, Project Page	
De novo protein design using geometric vector field networks	Oct. 2023
• Weian Mao[*], Muzhi Zhu[*], Zheng Sun[*], Shuaike Shen, Lin Yuanbo Wu, Hao Chen[†], Chunhua Shen[†] • ICLR 2024 spotlight, Paper, Project Page	

SELECTED PROJECTS

VITON: High resolution virtual try on based on generative models <i>Scientific research projects of Chu Kochen Honors College advised by Prof. Hao Chen</i>	2023
MiniSQL: A database system implemented in C++ <i>The course project of Database systems and concepts</i>	2023
Parallel optimization of big data pivots selection based on vectorization and OpenMP <i>High Performance Computing 101 summer course ACM-China IPCC 2022</i>	2022

SKILLS

Programming Skills: Python, C/C++, PyTorch, SQL

Language Skills: Mandarin (Native), English (TOEFL 104)